

Experience

Imaging scientist with 10 years of experience designing neuroimaging and neuropathology pipelines for observational studies, such as the Human Connectome Project (HCP), Dominantly Inherited Alzheimer Network Observational Study (DIAN-Obs), and the Harvard Aging Brain Study (HABS), and clinical trials, such as the DIAN Trials Unit (DIAN-TU)

2026-2028	Postdoctoral research fellow , Centile Bioscience Inc./Lund University
2023-2026	Postdoctoral research fellow , Harvard University/Massachusetts General Hospital - Designed a neuroimaging pipeline to quantify ^{18}F-MK-6240 tau PET off-target signals in extracerebral tissues and their spill-in into cerebral and cerebellar cortical depths and folds, which replaced a prior tissue-agnostic workflow and provided evidence of tissue-specific spill-in (HABS). - Other contributions: Designed neuroimaging pipelines for the first studies of ^{64}Cu-FBP8 fibrin PET in Alzheimer disease and ^{18}F-BCPP-EF mitochondrial complex I PET in Down syndrome. Led an autopsy validation study of amyloid-β and tau PET, plasma p-tau217, and the digital clock drawing test (HABS).
2017-2023	Graduate research fellow , Washington University in St. Louis - Designed a computer vision pipeline to quantify levels of amyloid-β deposition, tauopathy, microgliosis, and astrogliosis from digital histology, which replaced a prior semi-quantitative workflow and provided quantitative ground-truth evidence of localized clearance of amyloid-β in the first DIAN-TU clinical trial. - Other contributions: Led autopsy validation studies of amyloid-related imaging abnormalities (DIAN-TU); amyloid-β and tau PET (DIAN-Obs); and diffusion MRI (Knight ADRC). Led validation studies of the visual read procedure for ^{18}F-AV-1451 tau PET and the regional Centiloid method (DIAN-TU).
2012-2015	Undergraduate research fellow , Washington University in St. Louis/University of Oxford - Designed a computer vision pipeline to quantify the angles at which axonal fibers cross cortical surfaces from digital histology, which replaced a prior manual, paper-based qualitative workflow and provided quantitative ground-truth evidence of systematic errors in diffusion MRI fiber estimates from the HCP. - Other contributions: Designed a workflow to quantify three-dimensional axonal fiber orientations via passive tissue clearing, near-infrared fluorescent immunohistochemistry, multiphoton laser scanning microscopy, co-registration with MRI, and structure tensor analysis.

Education

2018-2023	Ph.D. in Imaging Science , Washington University in St. Louis
2018-2022	M.S. in Electrical Engineering , Washington University in St. Louis
2011-2015	B.A. in Biochemistry , Washington University in St. Louis
2007-2011	IB Diploma , Lee's Summit West High School

Funding

2026-2028	Data-Driven Life Science Postdoctoral Fellowship	\$213,584
2023-2026	PET/MR in Alzheimer's Disease and Related Dementias T32 Fellowship	\$181,032
2020-2021	Knight Alzheimer Disease Research Center T32 Fellowship	\$25,836
2018-2023	NSF Graduate Research Fellowship Program	\$138,000

Awards

2026	Human Amyloid Imaging Travel Scholarship	\$850
2025	Human Amyloid Imaging Travel Scholarship	\$600
2024	American Association of Neuropathologists Travel Award	\$1,000
2023	Poletsky Award	\$2,000

2021	McKelvey School of Engineering Dean's Scholarly Award	\$5,000
2020	Human Amyloid Imaging Travel Scholarship	\$1,000
2013	Howard Hughes Medical Institute Summer Undergraduate Research Fellowship	\$4,000
2011	National Merit Scholarship	\$2,500
2011	Missouri Boys State Scholarship	\$2,500
2010	Missouri Music Teachers Association State Winner in Piano Solo	\$0
2009	Missouri Music Teachers Association State Winner in Piano Solo	\$0

Publications

7 first-author and 27 co-authored publications, h-index: 29

Full list: https://scholar.google.com/citations?user=FiDgm&user=FiDgm_MAAAAJ

2026	Charles D Chen et al. "Identifying early changes in imaging, plasma, and digital cognitive biomarkers that correspond to Alzheimer disease pathology in the presence of autopsy-confirmed co-pathologies". In: <i>Alzheimer's and Dementia</i> (2026). https://doi.org/10.1002/alz.71092
2025	Austin McCullough* et al. "Regional effects of gantenerumab on imaging biomarkers in the DIAN-TU-001 trial". In: <i>Alzheimer's and Dementia</i> (2025). https://doi.org/10.1002/alz.70347
2025	Charles D Chen et al. "Immunohistochemical evaluation of a trial of gantenerumab or solanezumab in dominantly inherited Alzheimer disease". In: <i>Acta Neuropathologica</i> (2025). https://doi.org/10.1007/s00401-025-02890-7
2023	Charles Chen et al. "Comparing tau PET visual interpretation with tau PET quantification, CSF biomarkers, and longitudinal clinical assessment". In: <i>Journal of Alzheimer's Disease</i> (2023). https://doi.org/10.3233/JAD-230032
2023	Charles D Chen* et al. "Longitudinal head-to-head comparison of ¹¹ C-PiB and ¹⁸ F-florbetapir PET in a Phase 2/3 clinical trial of anti-amyloid- β monoclonal antibodies in dominantly inherited Alzheimer disease". In: <i>European Journal of Nuclear Medicine and Molecular Imaging</i> (2023). https://doi.org/10.1007/s00259-023-06209-0
2021	Charles D Chen et al. "Comparing amyloid- β plaque burden with antemortem PiB PET in autosomal dominant and late-onset Alzheimer disease". In: <i>Acta Neuropathologica</i> (2021). https://doi.org/10.1007/s00401-021-02342-y
2020	Charles D Chen et al. "Ante- and postmortem tau in autosomal dominant and late-onset Alzheimer disease". In: <i>Annals of Clinical and Translational Neurology</i> (2020). https://doi.org/10.1002/acn3.51237